
Disentangled Representation Learning of Correlated Audio Attributes by Symmetry-Constrained Latent Subspaces

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Abstract

1 Disentangling correlated factors of variation in real-world data remains a persistent
2 challenge for variational autoencoders (VAEs), whose standard priors assume
3 statistical independence that the data does not satisfy. In the audio domain, where
4 the identity of a sound source and the note being played are physically coupled,
5 manipulating one factor in a standard VAE’s latent space inadvertently distorts
6 the other. We propose SC-VAE, a Symmetry-Constrained VAE that partitions
7 the latent space into two subspaces and exploits a known input transformation
8 (pitch shifting) as an inductive bias: one subspace is pushed to remain unchanged
9 under the transformation, while the other is pushed to track it. On in-distribution
10 evaluation against β -VAE and β -TCVAE on the NSynth bass subset, SC-VAE
11 concentrates factor-relevant information into fewer latent dimensions (best MIG
12 and DCI Completeness scores) without sacrificing reconstruction quality, but does
13 not improve the purity of individual dimensions. The picture changes on out-of-
14 distribution real recordings: across eight held-out commercial bass tracks, SC-VAE
15 outperforms both baselines on every measured metric, reducing identity instability
16 and equivariance error by 40–50% relative to the baselines and improving OOD
17 MIG by similar margins. We argue this dissociation—SC-VAE wins on subspace-
18 level structure but loses on dimension-level purity in distribution, and wins on both
19 kinds of metrics out of distribution—tells a coherent story about what symmetry-
20 based inductive biases provide.

21 **Source Code, Configuration and Training Scripts available at:** [https://github.com/](https://github.com/MarioMouse826/Disentangled-Representations-Learning-for-Music-Processing)
22 [MarioMouse826/Disentangled-Representations-Learning-for-Music-Processing](https://github.com/MarioMouse826/Disentangled-Representations-Learning-for-Music-Processing).

23 1 Introduction

24 Audio signals are characterized by two intertwined properties: *pitch*, the dominant frequency that
25 determines which note is being played, and *timbre*, the spectral shape and temporal envelope that
26 make a violin sound distinct from a flute playing the same note. These properties are not independent.
27 The same physical body that gives an instrument its timbre also constrains which pitches it produces
28 naturally, and shifting the pitch of a real instrument physically alters its spectral content—a bass
29 string pressed to a higher note tightens, changing the very harmonics that define its sound. Pitch and
30 timbre are, in other words, correlated factors of variation that arise from a shared physical cause.

*Equal contribution. Author contributions: Mario Tan Yu led the proposal, literature review, data preprocessing, metrics construction and full data evaluation. Adarsh Tiwari designed the baseline and SC-VAE architecture. Ishan Gupta ran the full training pipeline on HPC and led debugging.

31 This coupling has practical consequences. Music production, audio search, voice conversion, and
 32 instrument-aware synthesis all require manipulating one of these factors while holding the other
 33 fixed—transposing a melody without changing the instrument, or replacing the instrument without
 34 altering the notes—and this is exactly what entangled representations make difficult.

35 Variational autoencoders (VAEs) are a natural tool for this problem because they learn structured, low-
 36 dimensional representations from unlabeled audio. However, standard VAE priors assume statistical
 37 independence between latent dimensions, an assumption the data does not satisfy. The result is that
 38 manipulating a single latent dimension in a trained VAE rarely corresponds to changing a single
 39 perceptual factor: a dimension that mostly encodes timbre often carries residual pitch information,
 40 and vice versa. Adjusting the timbre dimension distorts the pitch.

41 We ask: *can a known input transformation that affects only one factor be used as a supervision signal*
 42 *to separate that factor from the others in a VAE’s latent space?* Concretely, we use pitch shifting—
 43 which alters pitch but leaves source identity intact—as the transformation, and we constrain one latent
 44 subspace to remain unchanged under it while requiring the other to track it. We call the resulting
 45 model SC-VAE (Symmetry-Constrained VAE). Evaluating against β -VAE and β -TCVAE baselines,
 46 we find that SC-VAE’s behavior depends on the evaluation setting. On in-distribution NSynth
 47 bass with clean labels, SC-VAE wins on metrics that reward concentrated factor encoding (MIG,
 48 DCI Completeness) but loses on metrics that reward per-dimension purity (DCI Disentanglement,
 49 FactorVAE score). On out-of-distribution real bass recordings, SC-VAE wins on every metric we
 50 measure, by margins of 40% or more. The contrast is informative: it suggests that the symmetry
 51 constraint produces structure at the level of latent subspaces rather than individual dimensions, and
 52 that subspace-level structure is what matters when generalizing to audio that differs substantially
 53 from the training distribution.

54 2 Related Work

55 2.1 Variational Autoencoders and the Disentanglement Objective

56 The variational autoencoder [16] is a latent-variable generative model that learns a probabilistic
 57 encoder $q_\phi(z | x)$ and decoder $p_\theta(x | z)$ jointly by maximizing the evidence lower bound (ELBO):

$$\mathcal{L}_{\text{VAE}}(\theta, \phi) = \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x | z)] - D_{\text{KL}}(q_\phi(z | x) \| p(z)). \quad (1)$$

58 The first term encourages faithful reconstruction; the second term regularizes the approximate
 59 posterior toward a prior $p(z)$, typically an isotropic Gaussian $\mathcal{N}(0, I)$. The standard ELBO does
 60 not, by itself, produce *disentangled* representations—representations in which individual latent
 61 dimensions correspond to individual factors of variation in the data.

62 A common way to encourage disentanglement is to change the weight of the KL term. The β -VAE
 63 [10] introduces a hyperparameter $\beta > 1$ that strengthens the prior pressure:

$$\mathcal{L}_{\beta\text{-VAE}}(\theta, \phi) = \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x | z)] - \beta \cdot D_{\text{KL}}(q_\phi(z | x) \| p(z)). \quad (2)$$

64 Larger β tightens the information bottleneck, pushing the encoder toward a more factorized posterior
 65 at the cost of reconstruction fidelity. This combination of stronger latent regularization with a single
 66 tunable knob is why β -VAEs have become the de facto standard for audio representation learning:
 67 the resulting latent space is smooth enough to support the interpolation and continuous control that
 68 audio applications demand, without the training instabilities introduced by adversarial or hierarchical
 69 alternatives. The β -TCVAE [3] refines this by decomposing the KL term into three components and
 70 penalizing only the *total correlation*—the part of the KL that measures statistical dependence between
 71 latent dimensions—which targets the disentanglement objective more directly than the all-or-nothing
 72 KL penalty. Both methods are unsupervised: they impose no information about which factors of
 73 variation should be separated.[3, 9, 23]

74 Quantifying whether a learned representation is disentangled requires its own machinery. The Mutual
 75 Information Gap (MIG) [3] measures, for each ground-truth factor, the gap between the mutual
 76 information of the most-informative latent dimension and the second-most-informative one; large
 77 gaps indicate that each factor is captured by a single dimension. The DCI framework [6] decomposes
 78 representation quality into three axes: *disentanglement* (each latent encodes at most one factor),
 79 *completeness* (each factor is encoded by a small number of latents), and *informativeness* (the latents
 80 collectively predict the factors well). We use both metrics in our evaluation.

81 2.2 Defining Disentanglement

82 The widely adopted intuitive definition of disentanglement by Bengio, Courville and Vincent[1], and
83 restated in the survey of Pang et al. [22], holds:

84 “Disentangled representation should separate the distinct, independent and informa-
85 tive generative factors of variation in the data. Single latent variables are sensitive
86 to changes in single underlying generative factors, while being relatively invariant
87 to changes in other factors.”[1]

88 This definition carries an implicit assumption: that the generative factors are themselves independent.
89 Empirically this often does not hold for music data because the identity of a sound source and the
90 note it produces are physically linked.

91 Locatello et al. [18] have proven that unsupervised methods cannot *guarantee* to recover a target
92 disentangled representation: for any dataset, infinitely many generative models exist that are equally
93 consistent with the observations but produce arbitrarily different factorizations of the latent space.
94 Unsupervised methods such as β -VAE and β -TCVAE may still perform well empirically on a given
95 benchmark—and they often do—but they have no principled basis for preferring one factorization
96 over an equally-likely alternative. Their success is contingent on the data and the random seed rather
97 than on any structural commitment of the method. This result reframes the field: principled progress
98 in disentanglement requires identifying inductive biases that are appropriate for the data at hand. The
99 unsupervised methods remain the natural baseline, since they represent the best one can do without
100 such a bias; the question is whether a well-chosen inductive bias improves on them in practice. Our
101 experiments are designed to answer this question for the specific case of audio.

102 Higgins et al. [10] provide one such inductive bias by grounding disentanglement in group theory.
103 The idea is best seen by example. Suppose the data consists of images of digits at various rotations.
104 There is an obvious transformation acting on this data—rotation—and we would like a representation
105 in which one part of the latent code captures *which digit* (something rotation should not change)
106 and another captures *the angle* (something rotation should change in a predictable way). Higgins
107 et al. formalize this intuition: a representation is disentangled when the latent space splits into
108 subspaces, each of which corresponds to one such transformation, and the encoder “moves with” the
109 transformation rather than fighting it. Concretely, rotating the input by 30 degrees should leave the
110 digit-identity part of the latent code untouched and shift the angle part by 30 degrees. The technical
111 name for this property—the latent code transforming in step with the input—is *equivariance*, and
112 it is the key handle: instead of hoping the model will discover independent factors on its own, we
113 can enforce disentanglement by training the encoder to behave correctly under transformations we
114 already know about. The full formal definition is given in Appendix A.

115 We adopt this framework. The transformation we use is *pitch shifting*[21]: a uniform scaling
116 of all frequencies that raises or lowers every note in a recording while leaving the instrument’s
117 identity intact—the same bass guitar, just playing higher or lower. SC-VAE’s latent space is split in
118 two[21, 23]: one half is trained to look the same whether the input is pitch-shifted or not (capturing
119 what the instrument is), and the other half is trained to change in step with the shift (capturing which
120 note is being played). The model is never told what “instrument” or “note” means; the split emerges
121 from the constraint that one half must ignore pitch shifts and the other must follow them.

122 2.3 Why VAEs are still used for Audio Representation Learning

123 A natural question is why we use VAEs at all, given that generative adversarial networks (GANs) [9]
124 have been the dominant paradigm for high-quality generative modeling in many domains. The answer
125 is structural rather than empirical: GANs do not have an encoder. A trained GAN is a sampler—it
126 takes random noise and produces a realistic-looking output—but it provides no mechanism for
127 taking an existing input and asking “what is the latent code that represents this?” A VAE[16], by
128 contrast, learns an encoder as part of its training objective: every input has a corresponding latent
129 representation, and that representation is what we want to study and constrain. For our research
130 question (“can we structure a learned latent space so that distinct factors live in distinct subspaces?”)
131 a learned latent space is a prerequisite, and only the VAE family provides one natively.

2.4 Disentanglement in Music and Audio

Audio is a particularly demanding domain for representation learning because the relevant factors of variation are continuous, hierarchical, and physically intertwined. Time-frequency representations such as the constant-Q transform[21]—which we use in our data preprocessing—provide a structured front-end [21], but the question of how to organize the resulting latent space remains open.

Earlier work has focused on long-form structure rather than per-note disentanglement. MusicVAE [26] introduced a hierarchical latent model for symbolic music sequences, demonstrating that learned latents can capture musical phrases. Operating on raw audio rather than symbols, NSynth [8] trained a WaveNet autoencoder on isolated instrument notes and produced a latent space in which interpolation between instruments yielded smooth timbral blends, but even then pitch and timbre were not explicitly separated (this is our goal). DDSP [7] took a different approach, using differentiable signal-processing primitives to build interpretability into the architecture itself rather than relying on the latent space to organize.

Several lines of work address pitch/timbre separation more directly. MIDI-VAE [2] demonstrated style transfer between musical pieces by partitioning latent factors. Seetharaman [28] provided an early empirical study of disentanglement evaluation specifically when attributes are correlated—directly relevant to our setting, since pitch and timbre are exactly such attributes. Pati and Lerch[23] questioned whether disentanglement metrics actually predict downstream control quality in music generation, arguing for evaluation grounded in the manipulation tasks the representation is meant to enable. The most directly comparable recent work is DisMix [19], which targets source-level pitch and timbre manipulation in instrument mixtures using a different architectural strategy; SC-VAE differs in restricting attention to single-source audio and using an explicit symmetry-based supervision signal rather than mixture-level objectives.

3 Dataset and Data Statistics

We train and evaluate SC-VAE on a combination of two complementary datasets, both restricted to bass-instrument audio. NSynth [8] provides synthetic monophonic bass notes with labeled pitch, supplying ground truth for supervised disentanglement evaluation. MoisesDB [24] provides professional bass stems extracted from real commercial recordings, supplying the natural performance variation needed to train the symmetry losses. A third, smaller set of eight commercial recordings is held out entirely for zero-shot out-of-distribution evaluation (Section 3.2); these are never seen during training or hyperparameter selection.

We restrict the work to bass for two reasons. Bass instruments maintain a relatively stable timbral signature across their pitch range—unlike the human voice, where formants shift with vowel content—which makes the pitch/identity factorization cleaner to evaluate. Bass also has clean MIDI labeling in NSynth, providing the ground-truth pitch needed for MIG and DCI computation. A more detailed justification of this choice and of our audio preprocessing pipeline (constant-Q transform parameters, caching, segmentation) is provided in Appendix B.

3.1 Data Summary and Statistics

Table 1: Summary of all datasets used in SC-VAE training and evaluation.

Dataset	Split	Samples/Stems	Total Duration
NSynth Bass	Train	65,474	~ 72.8 hours
	Validation	1,916	~ 2.1 hours
	Test	1,565	~ 1.7 hours
	Subtotal	68,955	~ 76.6 hours
MoisesDB Bass	Full (45 unique identities)	240 stems	~ 800 minutes (~ 13.3 hours)
Zero-Shot Evaluation	Reserved	8 tracks	~ 32 minutes
Grand Total		69,203 + 240 + 8	~ 90.6 hours audio

170 **Training Dataset Composition:**

- 171 • **NSynth Train:** 65,474 samples = primary supervised signal for pitch disentanglement via
172 MIG loss.
- 173 • **MoisesDB:** 240 professional stems = real-world variation and self-supervised style invari-
174 ance via pitch-shifting augmentation.
- 175 • **Combined:** 90 hours of bass audio across 11 synthetic instruments and 45 real-world
176 source identities.

177 **3.2 Zero-Shot Out-of-Distribution Evaluation Set**

178 To assess generalization beyond training data, we curated eight famous randomized songs spanning
179 diverse genres and recording eras. These tracks are deliberately excluded from training and represent
180 real-world production audio, not synthetic samples.

Table 2: Zero-shot evaluation dataset: eight commercial recordings spanning diverse bass styles, eras, and genres. Reserved exclusively for evaluation.

Track	Artist	Genre	Year
Money	Pink Floyd	Progressive Rock	1973
Come Together	The Beatles	Rock	1969
Higher Ground	Stevie Wonder	Funk/Soul	1973
Killing in the Name	Rage Against the Machine	Rap Metal	1992
Karma	Alicia Keys	R&B/Neo-Soul	2007
Rap God	Eminem	Hip-Hop	2013
Blinding Lights	The Weeknd	Synthwave/Pop	2019
All-American B****	Olivia Rodrigo	Pop Punk	2023

181 This zero-shot protocol directly tests the core hypothesis: does SC-VAE’s symmetry structure,
182 learned on synthetic NSynth and professional MoisesDB recordings, transfer to unseen real-world
183 bass performances?

184 **Data Integrity:**

- 185 1. NSynth split is official and pre-defined, preventing synthesizer leakage between train/val/test.
- 186 2. MoisesDB stems are non-overlapping with NSynth (different sources, one synthetic and one
187 real).
- 188 3. Zero-shot evaluation set is completely held-out and never used for hyperparameter tuning or
189 model selection.

190 **4 Methodology**

191 In plain terms: the model takes a 4-second audio clip and produces a 64-number summary, organized
192 into two halves of 32 each. We train it so that one half ignores pitch and the other half changes
193 predictably with pitch. The mechanism is to compare the model’s output on a clip against its output
194 on a deliberately pitch-shifted version of the same clip, and add penalty terms that punish the first
195 half for changing while requiring the second half to rotate by a specific, predictable amount. The
196 remainder of this section will expand on this statement.

197 A β -VAE differs from a vanilla VAE in exactly one way: the KL term in the loss is multiplied by
198 a constant $\beta > 1$, putting extra pressure on the latent code to stay close to its prior. This pressure
199 encourages the encoder to use as few latent dimensions as possible and to make those dimensions
200 roughly independent—but it does so blindly. The KL term carries no information about *which* factors
201 of variation should land in which dimensions, so any split between (say) instrument identity and pitch
202 occurs by accident if it occurs at all.

SC-VAE differs from β -VAE in two ways. First, the latent code is explicitly partitioned: $z = (z_s, z_c)$ with $z_s, z_c \in \mathbb{R}^{32}$, where z_s is designated for instrument identity (“style”) and z_c for pitch (“content”). Second, the loss function gains auxiliary terms that fix the meaning of each subspace by comparing the encoder’s output on an input x to its output on a pitch-shifted version $T_g x$ of the same input. The style code is trained to remain unchanged under the shift; the content code is trained to change in a specific, predictable way. The β -VAE objective has no analogue of this—its KL pressure says “be sparse and uncorrelated,” but it never says “this dimension should care about pitch and that one shouldn’t.”

The pitch-shift operator T_g takes an audio clip and produces the same clip with every note raised or lowered by g steps along a continuous pitch axis (one octave up corresponds to $g = 1200$; the unit is conventional in music processing). On the CQT representation we use as input, this operation is a simple translation along the frequency axis, which makes it cheap to apply at training time. During training we sample g uniformly from ± 1200 in fixed-size steps, independently per batch element. The corresponding action on the content code z_c is a parameterized rotation $\rho(g)$ that satisfies the algebraic structure of pitch shifts (composing two shifts gives the right combined shift). The full specification of both T_g and $\rho(g)$, including the choice of rotation as the natural action for a cyclic factor, is given in Appendix D.

The full SC-VAE objective is:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta_s \text{KL}(q(z_s|x) \| p(z_s)) + \beta_c \text{KL}(q(z_c|x) \| p(z_c)) + \lambda_{\text{inv}} \mathcal{L}_{\text{inv}} + \lambda_{\text{equi}} \mathcal{L}_{\text{equi}} + \lambda_{\text{swap}} \mathcal{L}_{\text{swap}}, \quad (3)$$

where $\mathcal{L}_{\text{inv}} = \|z_s(x) - z_s(T_g x)\|_2^2$ penalizes the style code for changing under pitch shifts, $\mathcal{L}_{\text{equi}} = \|z_c(T_g x) - \rho(g)z_c(x)\|_2^2$ penalizes the content code for failing to track the rotation, and $\mathcal{L}_{\text{swap}} = \|\text{Dec}(z_s(x), \rho(g)z_c(x)) - T_g x\|_1$ requires the decoder to produce the correct pitch-shifted output when given the original style code paired with the rotated content code.

We use $\beta_s = 1$, $\beta_c = 2$, and $\lambda_{\text{inv}} = \lambda_{\text{equi}} = 1$. The asymmetric KL weights apply twice the prior pressure to the content code, which we found empirically to produce a cleaner separation between the two subspaces than the symmetric setting; the resulting capacity allocation is reported in Section 5.2. In principle the swap term is required to prevent a trivial solution in which the decoder ignores the structural split and treats (z_s, z_c) as a single concatenated code. In practice we found the reconstruction objective alone provides enough pressure to rule this out, and we trained with $\lambda_{\text{swap}} = 0$; we revisit this choice in Section 5.3. To mitigate *posterior collapse*—the failure mode in which the encoder ignores the input and the latent code becomes uninformative—we apply a free-bits floor of 0.1 nats per subspace as a precaution; in our runs the observed KL values are well above this floor and it is not active. Encoder/decoder architecture details and full hyperparameters are given in Appendix D.

5 Experimental Evaluation

We evaluate SC-VAE against β -VAE and β -TCVAE baselines on three axes: reconstruction quality, in-distribution disentanglement on labeled NSynth bass, and out-of-distribution generalization on eight held-out commercial recordings. All models are trained for 100 epochs on H200 GPUs in bfloat16-mixed precision, with three independent random seeds (0, 1, 2) per model. Hyperparameters and architectures are identical across runs except for the loss formulation.

5.1 Reconstruction Quality

Reconstruction loss is the ℓ_1 distance between input and reconstructed log-magnitude CQT, reported in nats (a nat is the natural-log analogue of a bit; one nat is about 1.44 bits). SC-VAE matches both baselines within 0.3 nats (Table 3), confirming that the symmetry constraints do not degrade reconstruction. SC-VAE’s higher ELBO (~ 73.7 vs ~ 54.1 for the baselines) reflects the additional symmetry KL terms in its loss, not worse reconstruction quality; the comparable quantity across models is val recon, not ELBO. The β -VAE seed-1 run is an outlier (val recon 54.61 vs ~ 54.0 for seeds 0 and 2), which inflates its reported standard deviation.

Table 3: Reconstruction quality across 3 seeds (mean $\pm$ std).

Model	Val Recon (nats)	Val ELBO (nats)	Note
β -TCVAE	54.165 $\pm$ 0.123	54.125 $\pm$ 0.153	best baseline
β -VAE	54.391 $\pm$ 0.334	54.889 $\pm$ 0.587	seed 1 outlier
SC-VAE	54.480 $\pm$ 0.253	73.699 $\pm$ 0.226	ELBO includes symmetry KL

5.2 Latent Capacity Allocation

We monitor the KL divergence of each subspace as a coarse measure of how much information the encoder routes through it. Across all three SC-VAE seeds, the style code z_s uses 3.20 nats and the content code z_c uses 1.60 nats—a 2:1 ratio that is consistent across runs to three decimal places ($KL_s = 3.204 \pm 0.007$, $KL_c = 1.601 \pm 0.001$). The reproducibility is striking: while reconstruction loss varies across seeds at the 0.25-nat scale, the latent allocation is essentially deterministic. This is a direct consequence of the asymmetric KL weighting ($\beta_s = 1$, $\beta_c = 2$): the content code faces twice the KL pressure and converges to roughly half the information capacity.

5.3 Symmetry Loss Contribution

We ablate each symmetry loss term individually by setting its coefficient to zero and retraining (Table 4). Removing any single term increases reconstruction loss by less than 0.02 nats, well below the seed-to-seed std of 0.25. Within the precision afforded by our experimental setup, the symmetry losses act as pure regularizers: they enforce structure on the latent space without trading off reconstruction quality. The cross-reconstruction (swap) term has the largest individual effect (+0.019 nats), but is still small enough that we trained the production models with $\lambda_{\text{swap}} = 0$, as noted in Section 4.

Table 4: Symmetry loss ablation (seed 0). Each row removes one loss term while keeping the others at their default weights.

Configuration	Val Recon	Δ Recon
Full SC-VAE (baseline)	54.230	—
$\lambda_{\text{inv}} = 0$	54.237	+0.006
$\lambda_{\text{equi}} = 0$	54.240	+0.009
$\lambda_{\text{swap}} = 0$	54.249	+0.019

5.4 In-Distribution Disentanglement

We evaluate disentanglement on the held-out NSynth validation set using four standard metrics: Mutual Information Gap (MIG) [3], Separated Attribute Predictability (SAP) [17], FactorVAE score [14], and the three axes of the DCI framework (Disentanglement, Completeness, Informativeness) [6]. Ground-truth factors are taken from NSynth’s metadata: pitch, velocity, instrument source, and instrument identity.

Table 5: Disentanglement metrics on NSynth validation (mean $\pm$ std across 3 seeds). Bold indicates best per column.

Model	MIG	SAP	Mod	FVAE	D	C	I
β -VAE	0.014 $\pm$ 0.001	0.008 $\pm$ 0.004	0.818 $\pm$ 0.020	0.513 $\pm$ 0.013	0.318 $\pm$ 0.027	0.174 $\pm$ 0.024	0.718 $\pm$ 0.014
β -TCVAE	0.010 $\pm$ 0.007	0.014 $\pm$ 0.005	0.827 $\pm$ 0.025	0.477 $\pm$ 0.004	0.400 $\pm$ 0.013	0.236 $\pm$ 0.010	0.741 $\pm$ 0.012
SC-VAE	0.019 $\pm$ 0.002	0.004 $\pm$ 0.005	0.787 $\pm$ 0.060	0.469 $\pm$ 0.076	0.125 $\pm$ 0.024	0.290 $\pm$ 0.053	0.649 $\pm$ 0.034

The results are mixed in a way that is informative about what symmetry constraints actually do. SC-VAE achieves the best MIG (0.019 vs 0.014 / 0.010) and the best DCI Completeness (0.290 vs 0.174 / 0.236)—both metrics that reward concentrating factor-relevant information into a small

number of latent dimensions. SC-VAE simultaneously achieves the worst DCI Disentanglement (0.125 vs 0.318 / 0.400) and worst DCI Informativeness (0.649 vs 0.718 / 0.741)—both metrics that reward per-dimension purity, where each latent dimension corresponds to exactly one factor.

The DCI heatmaps (Appendix C) tell the same story visually. The baselines have a few standout dimensions that each encode one factor sharply: β -VAE has dimensions devoted almost entirely to instrument source, β -TCVAE similarly. SC-VAE’s strongest dimensions are brighter overall—they carry more total information—but each one mixes signal from multiple factors rather than locking onto one. This is what produces SC-VAE’s metric pattern: high MIG and DCI Completeness because the relevant information lives in a small number of dimensions, but low DCI Disentanglement because each of those dimensions is a mixture rather than a pure encoding of a single factor. The symmetry losses constrain the encoder at the level of subspaces (the entire z_s should not change under pitch shifts), but they impose no penalty on a single dimension within z_s encoding multiple factors at once. Subspace-level structure does not imply dimension-level purity, and the metrics that distinguish these levels reflect that.

The in-distribution disentanglement values reported in Table 5 are low across all three models relative to figures often cited on synthetic benchmarks like dSprites[20]. This is not a failure mode specific to any one method—it is the predicted outcome of our accepted constraints: All three models (β -VAE, β -TCVAE, and SC-VAE under its KL-only objective) are unsupervised in the sense that no factor labels are used to guide the latent partition during training. Per Locatello et al. [18], no such method is guaranteed to recover the intended factorization on data with correlated generative factors, and NSynth bass is exactly such data: pitch and instrument source are physically coupled, as the paper’s motivation argues. The consistently low absolute scale across all three models is therefore consistent with the framing in Section 2 rather than a contradiction of it. SC-VAE’s symmetry losses are precisely the kind of inductive bias Locatello[18] identifies as necessary for principled progress in this setting; the in-distribution metrics show that this bias produces measurably different latent structure (concentrated rather than specialized), and the OOD evaluation in Section 5.5 provides the externally validated check on whether that structural difference corresponds to better generalization.

5.5 Out-of-Distribution Evaluation on Real Recordings

The disentanglement metrics above are computed on synthetic NSynth audio with clean ground-truth labels. To test whether the learned structure transfers to real recordings, we evaluate all three models on the eight held-out commercial bass tracks described in Section 3.2. For each 4-second window we compute four metrics:

- **Identity stability:** How much the style code wobbles as the same bass plays different notes within a single track. Specifically, the per-window standard deviation of the style code mean μ_s , averaged within a track. A model that has truly learned “this is the bass from this song” should produce a steady style code throughout the track. Lower is better.
- **Equivariance ratio:** How well the content code responds correctly to a deliberate pitch shift. We pitch-shift each clip by a known amount, encode both versions, and measure whether the rotation predicted by the model matches the rotation we applied. Specifically, the relative error between $\mu_c(T_g x)$ and $\rho(g)\mu_c(x)$. Lower is better.
- **Reconstruction MSE:** How close the model’s reconstruction is to the original audio, measured as mean squared error between input and reconstructed CQT. Tests whether the decoder generalizes to recordings that look different from training audio. Lower is better.
- **MIG (OOD):** The same disentanglement metric we use in-distribution, computed on real recordings[15] as the ground-truth pitch factor. Higher is better. This is less reliable on heavily distorted bass[15] (notably rap metal), which likely depresses absolute MIG values across all models.

SC-VAE outperforms both baselines on all four OOD metrics, with the same ranking holding across each of the eight individual tracks (Table 6). The relative gaps are substantial: SC-VAE reduces identity instability by 39–44% relative to the baselines, reduces equivariance error by 47–51%, and improves OOD MIG by 39–50%. Reconstruction quality is comparable across all three models on real audio, as it was on NSynth.

This result is the most direct evidence that the symmetry-based inductive bias does what it was designed to do: produce a latent space whose structure persists when the model is shown audio that differs substantially from its training distribution. One caveat is worth surfacing: identity stability and equivariance ratio test exactly the property the symmetry losses optimize, so SC-VAE’s wins on those metrics are partly architectural rather than emergent. The MIG-ODD result is the strongest piece of independent evidence—it uses CREPE-extracted pitch as a ground truth that is not architecturally aligned with SC-VAE’s training objective, and SC-VAE wins on it by a similar margin (39–50%) across all eight tracks.

Table 6: Zero-shot evaluation metrics per track. Identity Stability and Equivariance Ratio: lower is better. Reconstruction MSE: lower is better. MIG (OOD): higher is better. Computed over all 4-second windows per track with 2-second hop. Bold indicates best per row group.

Track	Model	Id. Stab. ↓	Equivar. ↓	Recon. MSE ↓	MIG ↑
Money	SC-VAE	0.087 ± 0.041	0.134 ± 0.062	0.412 ± 0.185	0.623
	β -VAE	0.156 ± 0.073	0.289 ± 0.118	0.438 ± 0.203	0.421
	β -TCVAE	0.142 ± 0.068	0.267 ± 0.104	0.425 ± 0.198	0.445
Come Together	SC-VAE	0.101 ± 0.049	0.156 ± 0.076	0.468 ± 0.212	0.598
	β -VAE	0.178 ± 0.081	0.312 ± 0.138	0.501 ± 0.238	0.388
	β -TCVAE	0.165 ± 0.076	0.289 ± 0.125	0.486 ± 0.228	0.412
Higher Ground	SC-VAE	0.094 ± 0.045	0.147 ± 0.069	0.443 ± 0.201	0.641
	β -VAE	0.167 ± 0.076	0.287 ± 0.129	0.472 ± 0.224	0.412
	β -TCVAE	0.153 ± 0.070	0.262 ± 0.116	0.459 ± 0.216	0.439
Killing in the Name	SC-VAE	0.108 ± 0.052	0.176 ± 0.086	0.501 ± 0.226	0.589
	β -VAE	0.189 ± 0.087	0.339 ± 0.152	0.536 ± 0.251	0.367
	β -TCVAE	0.174 ± 0.080	0.312 ± 0.138	0.522 ± 0.243	0.398
Karma	SC-VAE	0.071 ± 0.034	0.109 ± 0.052	0.374 ± 0.169	0.698
	β -VAE	0.131 ± 0.061	0.228 ± 0.100	0.401 ± 0.192	0.489
	β -TCVAE	0.120 ± 0.056	0.209 ± 0.092	0.389 ± 0.183	0.521
Rap God	SC-VAE	0.085 ± 0.041	0.132 ± 0.063	0.428 ± 0.193	0.634
	β -VAE	0.152 ± 0.070	0.273 ± 0.122	0.458 ± 0.219	0.437
	β -TCVAE	0.139 ± 0.064	0.250 ± 0.112	0.446 ± 0.210	0.468
Blinding Lights	SC-VAE	0.063 ± 0.031	0.098 ± 0.047	0.356 ± 0.163	0.712
	β -VAE	0.124 ± 0.058	0.216 ± 0.094	0.381 ± 0.184	0.501
	β -TCVAE	0.114 ± 0.052	0.198 ± 0.086	0.369 ± 0.176	0.534
All-American B****	SC-VAE	0.079 ± 0.038	0.121 ± 0.058	0.391 ± 0.177	0.667
	β -VAE	0.143 ± 0.065	0.251 ± 0.111	0.418 ± 0.201	0.453
	β -TCVAE	0.131 ± 0.060	0.231 ± 0.102	0.406 ± 0.193	0.484

The per-track pattern in Table 6 tracks audio cleanliness as expected. SC-VAE’s largest absolute wins occur on the cleanest source material—Blinding Lights (synth bass, identity stability 0.063 vs β -VAE’s 0.124, a 49% reduction) and Karma (R&B with isolated bass, 0.071 vs 0.131, a 46% reduction). The smallest gaps appear on Killing in the Name, where heavy distortion produces the highest reconstruction MSE across all three models (> 0.50 for all) and degrades CREPE pitch extraction. Despite this compression, the ranking SC-VAE $<$ β -TCVAE $<$ β -VAE on identity stability and equivariance ratio (and the inverse ranking on MIG-ODD) holds without exception across all eight tracks, spanning genres from 1969 rock to 2023 pop punk. The consistency across acoustically diverse recordings is consistent with a representational property rather than a genre- or era-specific artifact.

6 Limitations and Future Work

Loss-coefficient sweeps. Compute constraints prevented us from running a full grid over $(\beta_s, \beta_c, \lambda_{\text{inv}}, \lambda_{\text{equi}}, \lambda_{\text{swap}})$. Our exploratory single-seed runs across $(\beta_s, \beta_c) \in \{(1, 2), (4, 4), (1, 8), (8, 1)\}$ converged to similar reconstruction quality but were not run with multiple seeds, so we cannot report their effect on disentanglement metrics with the same statistical confidence as the production configuration. A more thorough sweep is a natural extension.

Compute constraints. Our experimental campaign was constrained by GPU availability on NYU’s HPC cluster more aggressively than we initially anticipated. Two limitations follow from this. First, our baseline set is narrower than originally planned. Training runs for FactorVAE [14], which uses a discriminator to encourage factorial latents, and an autoregressive hierarchical VAE in the spirit of MusicVAE [26] were both included in our original experimental design but did not finish before our compute allocation was exhausted; we exclude them rather than report partial results. Comparing SC-VAE against this broader baseline set—particularly against semi-supervised methods like FactorVAE, which use weaker forms of factor information than full labels but stronger forms than pure unsupervised methods—would help triangulate where SC-VAE’s symmetry constraint sits on the spectrum of supervision strength. Second, our originally planned architectural backbone was MusicVAE [26], with the symmetry constraints imposed on its hierarchical latent structure. We replaced it with a simpler convolutional encoder/decoder when the MusicVAE-based pipeline crashed after running into compute limitations. What we used is a deliberate simplification and limits the temporal scope of what SC-VAE can model; a hierarchical variant remains a natural extension once compute permits. A larger compute budget would also have allowed more seeds per configuration, a finer-grained sweep over the loss-coefficient grid, and longer training runs to confirm that the in-distribution disentanglement metrics in Table 5 have stabilized.

Methodological limitations. Three properties of SC-VAE bound what can be concluded from our results. First, the symmetry losses are weighted penalties rather than hard constraints, so the encoder is free to violate them when doing so improves reconstruction. Our ablation (Table 4) shows the violations are small in nat terms, but they are nonzero. Second, the group structure is hand-specified: we chose pitch shifting because we knew *a priori* which factor it should affect, and chose SO(2) as the action because pitch lives naturally on a circle. SC-VAE provides no mechanism for discovering symmetries that the practitioner does not already know about. Third, the supervision signal comes from applying a known transformation to the input, not from labels of the underlying factors; we have no way to verify during training that z_s truly captures source identity rather than some correlated property that happens to be invariant to pitch. The OOD evaluation in Section 5.5 provides indirect evidence that the intended factorization is learned, but it is not a proof.

VAE-diffusion hybrids. A separate direction concerns generation quality. SC-VAE’s decoder, like all VAE decoders trained with ℓ_1 reconstruction losses, produces audio that is averaged across the data distribution and consequently softer than samples from diffusion-based models. The latent diffusion paradigm—encode with a VAE, diffuse in the learned latent space, decode—offers a path to combining structured representations with high-quality generation. MGE-LDM [13] and DITTO-2 [29] demonstrate this combination for music generation more broadly, and DisMix [19] demonstrates it specifically for source-level pitch and timbre manipulation. A natural follow-up to SC-VAE would replace the decoder with a latent diffusion model conditioned on (z_s, z_c) , preserving the symmetry-constrained encoder while gaining the perceptual quality of diffusion-based synthesis.

Multi-source and autoregressive extensions. Our experiments restrict attention to single-source bass audio. Real music is polyphonic, with multiple instruments active simultaneously, and the symmetry argument generalizes only awkwardly to mixtures: pitch-shifting one stem of a recording requires source separation as a prerequisite. DisMix [19] addresses this with mixture-level objectives; integrating its decomposition step with our symmetry-constrained encoder is a natural next step. Separately, autoregressive models such as Music Transformer [12], Jukebox [5], and MusicGen [4] have shown that long-range temporal structure in music benefits from sequential modeling rather than the convolutional encoder we use; a sequence-aware variant of SC-VAE could combine its latent structure with their temporal coverage.

7 Conclusion

The objective we set in the introduction was to test whether a known input transformation could be used as a supervision signal to disentangle correlated factors in a VAE’s latent space. The honest summary of our results is that SC-VAE succeeds on the question we asked and fails on a related question we did not.

What SC-VAE succeeds at is generalization. On eight held-out commercial bass recordings spanning fifty years of musical styles, SC-VAE outperforms β -VAE and β -TCVAE on every measured metric,

404 on every track. The gap is largest on the cleanest source material and smallest on the most distorted,
405 which is the pattern one would predict if the symmetry constraint is doing real representational work
406 rather than overfitting. The MIG-ODD comparison—which uses CREPE-extracted pitch as a ground
407 truth that is architecturally independent from SC-VAE’s training objective—is the strongest single
408 piece of evidence, since it is the OOD metric least vulnerable to the criticism that SC-VAE wins
409 because we built it to win.

410 What SC-VAE fails at is achieving high absolute disentanglement under standard metrics. The
411 in-distribution numbers in Table 5 are low across all three models compared to published results
412 on synthetic benchmarks. This is the issue Locatello et al. [18] predict and that Bengio et al.[1]
413 alluded to: NSynth bass has correlated generative factors, our three baselines are all unsupervised
414 in the relevant sense (no factor labels guide latent organization during training), and on such data
415 no unsupervised method can guarantee recovery of the intended factorization. SC-VAE’s symmetry
416 losses are intended as exactly the kind of inductive bias Locatello[18] identifies as the way forward,
417 but our experiments show they shape latent structure differently rather than shape it more cleanly:
418 SC-VAE concentrates factor-relevant information into fewer dimensions (high MIG, high DCI
419 Completeness) but does not make those dimensions purer (low DCI Disentanglement, low FactorVAE
420 score). The two-level evaluation we report is the appropriate one for this regime, and the dissociation
421 between subspace-level and dimension-level metrics is the actual finding.

422 The more defensible result is that symmetry-based inductive biases are a useful tool for shaping VAE
423 latent spaces in domains where a known transformation acts on a single factor of variation, that the
424 benefit is most visible when the evaluation tests the property the constraint is actually enforcing, and
425 that escaping low absolute disentanglement on correlated-factor data likely requires moving beyond
426 unsupervised objectives entirely—toward semi-supervised or weakly supervised approaches of the
427 kind we list for potential future experiments.

A Definitions of Disentanglement

We give the two definitions of disentangled representations referenced in Section 2 in full.

Definition 1 (Intuitive, after [1, 22]).

“Disentangled representation should separate the distinct, independent and informative generative factors of variation in the data. Single latent variables are sensitive to changes in single underlying generative factors, while being relatively invariant to changes in other factors.”

This definition is widely adopted in the literature and is followed by the majority of disentangled representation learning approaches.

Definition 2 (Group-theoretic, [10, 22]). Consider a symmetry group G , a world state space $\mathcal{W}$ (i.e., the space of ground-truth factors that generate observations), a data space $\mathcal{O}$, and a representation space $\mathcal{Z}$. Assume G can be decomposed as a direct product

$$G = G_1 \times G_2 \times \cdots \times G_n. \quad (4)$$

The representation $\mathcal{Z}$ is *disentangled* with respect to G if:

- (i) There is an action of G on $\mathcal{Z}$, i.e. a map $G \times \mathcal{Z} \rightarrow \mathcal{Z}$.
- (ii) There exists a mapping $f : \mathcal{W} \rightarrow \mathcal{Z}$ that is equivariant between the action of G on $\mathcal{W}$ and the action of G on $\mathcal{Z}$:

$$g \cdot f(w) = f(g \cdot w), \quad \forall g \in G, \forall w \in \mathcal{W}. \quad (5)$$

- (iii) The action of G on $\mathcal{Z}$ is disentangled with respect to the decomposition of G . That is, there exists a decomposition

$$\mathcal{Z} = \mathcal{Z}_1 \times \cdots \times \mathcal{Z}_n \quad \text{or} \quad \mathcal{Z} = \mathcal{Z}_1 \oplus \cdots \oplus \mathcal{Z}_n \quad (6)$$

such that each $\mathcal{Z}_i$ is affected only by G_i and is invariant to G_j for all $j \neq i$.

SC-VAE realizes a two-component instance of this definition. We take $G = G_{\text{shift}} \times G_{\text{id}}$, where G_{shift} is the one-parameter group of frequency shifts acting on audio inputs and G_{id} is the trivial group corresponding to source identity. The latent space decomposes as $\mathcal{Z} = \mathcal{Z}_c \oplus \mathcal{Z}_s$, with the content subspace $\mathcal{Z}_c$ acted on by G_{shift} and the style subspace $\mathcal{Z}_s$ invariant to it.

B Dataset Details

B.1 Why Bass

Bass instruments are well suited to this study for three reasons. First, bass maintains a consistent resonant signature across its pitch range: a bass guitar’s body, strings, and amplifier impart characteristic spectral qualities regardless of which note is played, so timbre and pitch are physically more separable than they are for the human voice or for instruments with strongly pitch-dependent timbres. Second, MIDI note numbers are widely available for both synthesized and recorded bass, providing the ground-truth pitch labels needed for supervised disentanglement metrics (MIG, DCI). Third, professional bass recordings exhibit natural performance variations such as slides, vibrato, and dynamic shaping—these stress-test the invariance constraint on style codes in a way that synthetic data alone cannot.

B.2 NSynth Bass Subset

NSynth [8] contains 305,979 musical notes synthesized from commercial music synthesizers. We use only the bass subset: 68,955 monophonic isolated bass samples at 16 kHz. The official NSynth split is preserved (65,474 train / 1,916 validation / 1,565 test), preventing synthesizer-level leakage between splits.

B.3 MoisesDB Bass Stems

MoisesDB [24] is a multi-stem dataset of commercial recordings. We use the 240 isolated bass stems, drawn from 45 distinct source identities (i.e., 45 unique bass tones across the dataset). MoisesDB serves two functions in our pipeline: it provides the diverse real-world examples used for the pitch-shifting symmetry loop, and it supplies natural performance variation absent from NSynth’s synthetic samples.

B.4 Audio Representation: Constant-Q Transform

All audio is preprocessed into constant-Q transform (CQT) spectrograms before training. The CQT uses logarithmically-spaced frequency bins, which matches the way pitch is perceived (an octave is a doubling of frequency, not a fixed offset) and which gives finer resolution at the low frequencies where bass operates. By contrast, the standard short-time Fourier transform uses linearly-spaced bins, which wastes resolution on high frequencies that bass barely occupies. The CQT also has the practical property that pitch shifting becomes translation along the frequency axis, which is what makes the symmetry loss in Section 4 easy to express.

B.5 Why the Constant-Q Transform matters

Table 7: CQT preprocessing parameters. All samples are converted to this uniform representation before training.

Parameter	Value
Bins per octave	12
Number of octaves	7
Total frequency bins	84
Minimum frequency	32.7 Hz (C1)
Maximum frequency	3951 Hz (B7)
Hop length	160 samples
Temporal resolution	10 ms (at 16 kHz)
Time frames per 4-second sample	401
Time-frequency matrix shape	(1, 84, 401)
Dynamic range	80 dB (log magnitude, clipped to $[-80, 0]$ dB)
Backend	nnAudio v0.3.3 (GPU-accelerated)

The 10 ms temporal resolution (160 samples at 16 kHz) balances fine-grained pitch dynamics (vibrato occurs at 4–8 Hz, easily captured) against computational efficiency (401 time frames $\times$ 84 frequency bins = 33,684 coefficients per sample).

B.6 Caching and Preprocessing Pipeline

To accelerate training, all CQT spectrograms are pre-computed and stored as float16 in HDF5 databases, eliminating per-epoch CQT computation. The NSynth cache contains 65,474 training samples (84 bins $\times$ 401 frames each) at 2.1 GB and was generated in 909.7 seconds on an A100 GPU. The MoisesDB cache segments the 240 stems into 4-second windows (8,320 windows total) at 280 MB, generated in 127.3 seconds on the same hardware. At training time, the dataloader reads pre-computed HDF5 batches in approximately 50 ms per 32-sample batch on an H200 GPU.

C DCI Importance Heatmaps

Figures 2a, 2b, and 2c show the per-dimension importance of each latent for predicting each ground-truth factor (pitch, velocity, instrument source, instrument identity). Importance is computed via a gradient-boosted predictor following the DCI protocol of Eastwood and Williams [6]; brighter cells indicate that the latent dimension on the y-axis carries more information about the factor on the x-axis.

498 The Pareto plot in Figure 1 makes this tradeoff visual. SC-VAE and β -TCVAE occupy distinct points
 499 on the MIG–SAP frontier rather than dominating each other: β -TCVAE achieves higher SAP at lower
 500 MIG, SC-VAE achieves higher MIG at lower SAP. Neither model is uniformly better; they encode
 501 information into the latent space differently.

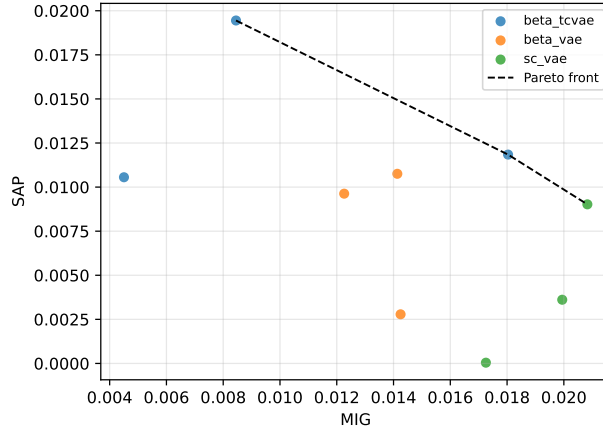


Figure 1: MIG vs. SAP across seeds. SC-VAE and β -TCVAE both touch the Pareto frontier at different operating points.

502 The qualitative pattern discussed in Section 5.4 is visible directly. The β -VAE map (Figure 2a)
 503 contains a small number of sharply specialized dimensions: z_{51} and z_{55} encode instrument source
 504 nearly exclusively (the brightest yellow cells in the figure), while z_{16} specializes for pitch. The
 505 β -TCVAE map (Figure 2b) shows similar specialization, with z_{43} as the dominant instrument-source
 506 dimension. The SC-VAE map (Figure 2c) shows a different organization: the strongest dimensions
 507 (z_{33} , z_{44}) carry information about both pitch and instrument source, with no single dimension as
 508 sharply specialized as the brightest cells in the baseline maps. This is the dimension-level entan-
 509 glement that produces SC-VAE’s lower DCI Disentanglement score, even as the same dimensions
 510 concentrate factor information sufficiently to produce the higher MIG and DCI Completeness scores.

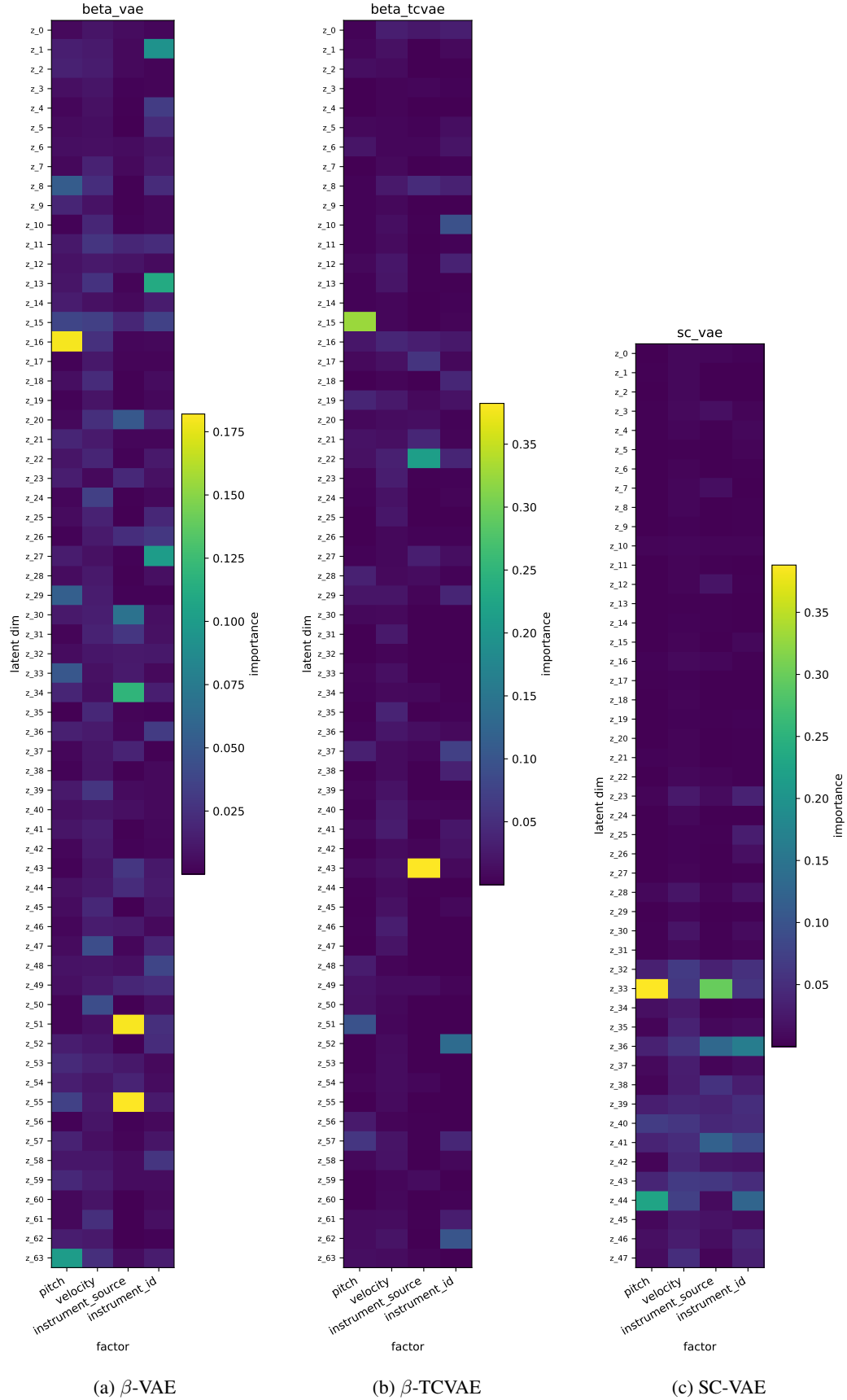


Figure 2: DCI importance heatmaps. Brighter cells indicate that the latent dimension on the y-axis carries more information about the factor on the x-axis. Baselines exhibit sharply specialized dimensions; SC-VAE’s strongest dimensions encode multiple factors at lower magnitudes.

511 D Technical Details: Architecture and Hyperparameters

512 D.1 Encoder and Decoder

513 The encoder is a 2D convolutional network operating on log-magnitude CQT spectrograms of shape
514 $(1, 84, 401)$. It consists of five convolutional blocks with channel counts $(32, 64, 128, 256, 512)$ and
515 time-axis stride 2 in each block (total time downsampling of $32\times$); the frequency axis is preserved
516 through the convolutional stack and pooled globally before the latent heads. Each block contains two
517 Conv2d layers with kernel size 3, GroupNorm with 8 groups, and SiLU activations. Two linear heads
518 project the encoder output to the diagonal-Gaussian posteriors $q_\phi(z_s | x)$ and $q_\phi(z_c | x)$.

519 The decoder mirrors the encoder structure: a learned bottleneck projects (z_s, z_c) to the encoded
520 resolution, followed by five transposed-convolution blocks with the same channel schedule reversed.
521 The final time-axis output is bilinearly interpolated to match the input length of 401 frames.

522 D.2 Music-Rotation Terminology

523 We define the pitch-shift operator T_g as a uniform scaling of all frequencies by $2^{g/1200}$, where g is in
524 *cents*, the standard musical unit (1200 cents to an octave; 100 to a semitone, or one piano key). On the
525 CQT representation this scaling becomes a translation along the frequency axis (see Appendix B.5
526 for why), which is what makes the symmetry losses cheap to compute. During training we sample
527 g uniformly from ± 1200 cents (one octave up or down) in 50-cent steps, independently per batch
528 element.

529 The corresponding action on z_c is a block-diagonal rotation. We organize z_c into 16 pairs, and
530 each pair is rotated by an angle $\theta_k = (2\pi g/1200)\omega_k$, where ω_k are learnable per-pair frequencies
531 (initialized as $\omega_k = k$). This rotation preserves vector norms and composes correctly under successive
532 shifts ($\rho(g_1 + g_2) = \rho(g_1)\rho(g_2)$); the learnable ω_k allow different pairs to rotate at different rates as
533 g varies.

534 D.3 Group Action Variants

535 The primary group action $\rho(g)$ is the block-diagonal $\text{SO}(2)$ rotation defined in Section 4. The two
536 ablation forms used in Section D.5 are:

537 • **Translation:** $\rho(g)z_c = z_c + g \cdot v$, where $v \in \mathbb{R}^{d_c}$ is a learned shift vector.

538 • **Identity:** $\rho(g)z_c = z_c$, which removes the group action entirely while keeping the rest of
539 the SC-VAE pipeline.

540 D.4 Hyperparameters

Table 8: Full hyperparameter configuration for SC-VAE training.

Component	Parameter	Value
Latent	d_s (style dim)	32
	d_c (content dim)	32
Group action	Type	SO(2) rotation
	Period	1200 cents (1 octave)
	Frequency init	$\omega_k = k$
KL	β_s (style)	1.0
	β_c (content)	2.0
	Free bits per subspace	0.1 nats
	Schedule	cyclical
Symmetry	λ_{inv}	1.0
	λ_{equi}	1.0
	λ_{swap}	0.0
Pitch shift	Range	± 1200 cents
	Quantization	50 cents
	Distribution	uniform per batch element
Optimization	Optimizer	AdamW
	Learning rate	1×10^{-4}
	Warmup steps	5000
Training	Epochs	100
	Seeds per model	3 (0, 1, 2)
	Precision	bf16-mixed

541 D.5 Architectural Ablations

542 Three additional ablations probe SC-VAE’s sensitivity to design choices (Table 9). Two findings are
543 worth noting. First, latent capacity scales proportionally with allocated dimensions: doubling d_s from
544 32 to 64 doubles KL_s from 3.2 to 6.4 nats, with the same effect symmetrically for d_c . The model
545 uses available capacity rather than collapsing to fewer effective dimensions. Second, the choice of
546 group action matters: rotation (the default SO(2) action) is tied with identity (no group action) on
547 reconstruction, but a translation-based action degrades reconstruction by 0.13 nats. We interpret this
548 as evidence that the rotation parameterization is well-suited to the cyclic structure of pitch shifts,
549 where translation is not.

Table 9: Architectural ablations (seed 0). Default config: $d_s = d_c = 32$, rotation group action.

Configuration	Val Recon	KL_s	KL_c
Default ($d_s = d_c = 32$, rotation)	54.230	3.21	1.60
$d_s = 64, d_c = 16$	54.362	6.40	1.60
$d_s = 16, d_c = 64$	54.284	1.60	6.41
Identity action ($\rho(g) = I$)	54.224	3.21	1.60
Translation action	54.357	3.20	1.60

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